
THE EVOLUTION OF MACHINE LEARNING AND AI

Week 1 Quiz Summary

The Shape of the Field: everything the quiz can ask, in plain English

How to use this page. Week 1 is examined in the **Week 6 quiz** (50% of the module). The quiz is **multiple choice**. Everything it can ask about Week 1 is in the **eleven points** below, one per part of the lecture.

Read the boxes (about 20 minutes). Do the questions (about 20 minutes). Mark them. Re-read only the points you got wrong.

If you remember one thing. AI's history goes round in circles. Something real works, people promise far too much, the money stops, the work carries on quietly under a new name, and then it happens again. Two of those cold spells are called **AI winters**.

1 Part 1. The eleven points

1.1 The pattern that repeats

Every climb starts with something that genuinely works. Then the promises run ahead of the results. Then the money and the attention collapse: an **AI winter**. Then people keep working quietly, and the field comes back under a new name. This has happened twice: **1974 to 1980**, and **1987 to 1993**. From the outside, a quiet field and a dead field look the same. Winters punish **popularity**, not **truth**.

1.2 Where AI came from (up to 1950)

AI started as an **argument**, not an invention. The argument: if the brain is doing a kind of computation, then a machine could in principle do the same thing. In 1950 **Turing** turned the question “can machines think?” into something you can actually test, the **imitation game**, and answered nine objections to it before anyone raised them.

1.3 The naming (1956)

At the **Dartmouth** workshop the field got its name, “artificial intelligence”, and its habit: prove progress by **building something and testing it**. It also got the split that has organised everything since. Two families: **symbolic**, meaning write the rules down, and **connectionist**, meaning learn from examples.

1.4 The golden years (1956 to 1969)

Everything worked, in the shallow end. Small versions of hard problems were solved, and people concluded the big versions would follow shortly. They did not. The lesson: **doing well on a toy version tells you very little about the real thing**. The reason it fails is the **combinatorial explosion**: as a problem gets bigger, the number of possibilities grows so fast that searching through them stops working.

1.5 The perceptron and the first winter (1958 to 1980)

Rosenblatt’s **perceptron** (1958) was one artificial neuron that learned from examples, and it could only draw a straight line. In 1969 **Minsky and Papert** proved it could not learn **XOR**, four points no straight line can separate. The proof was correct, but it was about the **one-layer** version, and it was read as a verdict on the whole idea. So the lesson is a habit: when you read a criticism, ask **which version does it actually apply to, and is there a way round?** Here there was a way round, backpropagation, and it took **seventeen years** to arrive.

1.6 Expert systems and the bottleneck (1980 to 1987)

The symbolic family sold itself commercially by writing experts’ rules into software. It hit the **knowledge acquisition bottleneck**: experts cannot say out loud most of what they know. Machine learning’s answer, and its bet, is that this unspoken knowledge can be **picked up from examples** instead of being interviewed out of people.

1.7 The second winter and the quiet revolution (1987 to 2011)

The expert-systems industry collapsed in 1987 and the field **changed its name to machine learning**. That was partly camouflage and partly honest: the work really had shifted. Three shifts: **rules to data, logic to probability, demonstrations to benchmarks**. By 2005 you used learned models every day, in spam filters and search, while believing AI had failed. That is the **AI effect**: once a machine can do something, we stop calling it intelligence, so “real AI” keeps meaning whatever machines cannot do yet.

1.8 The big bang (2012)

Deep learning took off only when **three supplies existed at the same time: architecture** (backpropagation, 1986; convolution, 1979), **compute** (CUDA, which turned gaming graphics cards into general-purpose number crunchers, 2007) and **data** (ImageNet, 14 million labelled photographs, 2009). **AlexNet** (2012) used old ideas on new supply. Almost nothing about the maths was new. **No single ingredient produces the result; the jump comes from the combination.**

1.9 Deep Blue against AlphaGo

Deep Blue (1997) beat the world chess champion with hand-written rules and brute-force search of a closed world, and it could do **nothing else**: zero transfer. **AlphaGo** (2016) **learned** how to judge a position and improved by playing itself, and its successor learned three different games with one method. Same kind of headline, completely different kind of achievement.

1.10 The language era and the four walls (2017 to now)

Attention (2017) plus a lot of scale gave us the language era and the chat box. **Scaling worked.** It is now running into four walls: **cost** (hundreds of millions for one frontier run), **energy** (grid-scale electricity, now a policy question), **data** (the internet's text, largely used up) and **diminishing returns**. Two turns follow: toward **science** and toward **lean**.

1.11 The module thesis

One sentence to carry for twelve weeks: **scaling worked, scaling is hitting walls, the field is turning.** And the position to hold on what comes next: **calibrated uncertainty, argued.** Anyone who is certain in either direction is selling something.

2 Part 2. The seven questions for any AI headline

You may be asked to **apply** these, so read each one as a test you could run.

#	Question
1	What did it actually do, and in what world: a swimming pool, or the ocean?
2	Would it survive a benchmark a rival could run?
3	Whose supply made it possible, and will that supply last?
4	Is this a new engine, or a new interface on an old engine?
5	Who is speaking, and what are they selling?
6	Which version of the idea does the criticism actually reach?
7	Is this brute force, science, or lean?

3 Part 3. Twenty practice questions

Same format as the quiz: one right answer each. Answers on the last page.

1. A company demonstrates a system that plans deliveries perfectly for a simulated town of 40 houses, and says a national roll-out is next. Which Week 1 lesson applies most directly?

- A. Success on a toy version is weak evidence about the full problem
- B. Winters punish popularity, not truth
- C. The knowledge acquisition bottleneck will stop them
- D. The AI effect: people will redefine the achievement away

2. A commentator writes: “Neural networks were proved useless in 1969.” What is the best response?

- A. Reply that proofs do not apply to modern hardware
- B. Point out that the proof was later shown to be wrong
- C. Ask which version the proof applies to: it was about the single-layer perceptron, and there was a way round
- D. Agree, because the proof was correct

3. A research area loses its funding and disappears from the news, but a handful of labs keep publishing. On Week 1’s account, what should you conclude?

- A. The ideas have been refuted, or the funding would have continued
- B. The field is dead, because from outside it looks dead
- C. The remaining labs are working on something unrelated
- D. Nothing about whether the ideas are sound: winters punish popularity, not truth

4. A vendor says its new model is a fundamental scientific breakthrough. Applying the headline questions, which single question most often deflates that claim?

- A. Is this an engine or an interface?
- B. Whose supply made this possible, and is that supply durable?
- C. Would it survive a benchmark a rival can run?
- D. Who is speaking, and what are they selling?

5. A system beats the world champion at one board game using hand-written rules and exhaustive search, and cannot play anything else. Which 1997 system does it resemble, and why?

- A. Deep Blue: brute-force search in a closed world, with zero transfer
- B. AlexNet: because it needed large compute
- C. AlphaGo: because it beat a human champion
- D. The perceptron: because it used a fixed rule

6. Backpropagation was published in 1986, yet deep learning did not take off for another 26 years. What is the best explanation?

- A. The 1986 method contained an error that was fixed in 2012
- B. The other two supplies, data and compute, had not yet arrived
- C. Nobody had thought to stack more layers
- D. Funding was unavailable until after the second winter

7. A team reports a model that is perfect on the examples it studied and much weaker on examples it has never seen. In Week 1’s terms, what has happened?

- A. It memorised the examples rather than learning the pattern, which is what a large model does with a small dataset
- B. The test examples were harder than the training ones
- C. The model needed more compute
- D. The benchmark was badly designed

8. Two labs each announce an advance. One publishes a benchmark result that rivals can reproduce; the other shows a polished video. What does Week 1 say about the difference?

- A. The video is stronger, because demonstrations persuade funders
- B. Neither counts unless the code is open source
- C. The reproducible benchmark is the stronger claim, because the field advances by tests rivals can run
- D. Both are equally strong until peer review

9. “Chess engines aren’t really intelligent, it’s just search.” This remark is an example of:

- A. The AI effect: once a machine can do something, we stop calling it intelligence
- B. The knowledge acquisition bottleneck
- C. The combinatorial explosion
- D. An AI winter beginning

10. A consultancy tries to capture a senior radiologist’s judgement as a set of if-then rules and finds she cannot explain most of what she does. What is this, and what is machine learning’s alternative?

- A. Label bias; hire a second radiologist
- B. The combinatorial explosion; use a faster search
- C. The AI effect; redefine the task
- D. The knowledge acquisition bottleneck; learn the judgement from many labelled examples instead

11. Frontier training runs now cost hundreds of millions, and the internet’s text is largely consumed. Which walls are these, and what follows in Week 1’s account?

- A. Cost and energy; the field turns toward bigger datasets
- B. Energy and returns; the field turns toward regulation
- C. Data and returns; the field enters a third winter
- D. Cost and data; the field turns toward science and toward lean

12. A product launches a chat interface on top of an existing model and calls it a breakthrough. Which headline question exposes this fastest?

- A. Pool, or ocean?
- B. Brute force, science, or lean?
- C. Engine, or interface?
- D. Which version does the criticism reach?

13. Which statement is the defensible position on what AI will do by 2030, according to Week 1?

- A. Scaling will continue to deliver, because it always has
- B. It cannot be discussed until the results are in
- C. Calibrated uncertainty, argued from evidence: anyone certain either way is selling something
- D. Progress will stall, because the walls are absolute

14. A system answers questions by matching them against a hand-built set of logical rules. Which family is it, and what is the other family’s approach?

- A. Connectionist; the symbolic family uses probability instead
- B. Symbolic; the connectionist family learns from examples instead
- C. Symbolic; the other family uses faster hardware
- D. Connectionist; the other family writes the rules down

15. Two ingredients of 2012 arrived before the third. Which ordering is right, and why does it matter?

- A. They all arrived in 2012, which is why the jump happened then
- B. Architecture (1986), then compute (2007), then data (2009); the jump came only when all three coexisted
- C. Compute, then data, then architecture; hardware always leads
- D. Data, then architecture, then compute; data is always the binding constraint

16. A benchmark is run in a simulator with perfect information and no noise. Which headline question should you ask first?

- A. Whose supply made it possible?
- B. Brute force, science, or lean?
- C. Who is speaking, and what are they selling?
- D. What did it actually do, and in what world: pool or ocean?

17. Which sentence best captures why AlexNet mattered in 2012?

- A. Old ideas finally met the data and compute they needed
- B. Researchers stopped using benchmarks
- C. Neural networks were invented
- D. A new learning algorithm replaced backpropagation

18. A colleague argues: “We should not collect any more data; the model is already the most flexible architecture available.” Which Week 1 idea most directly challenges this?

- A. No single ingredient carries the result; the jump comes from the combination
- B. Winters punish popularity, not truth
- C. The AI effect
- D. Toy success is weak evidence

19. In about 1990 many researchers stopped writing “AI” on their grant applications and wrote “machine learning” instead. What does Week 1 say this signalled?

- A. Only camouflage; the work was unchanged
- B. A genuinely new field with no connection to AI
- C. Both camouflage after the collapse and an honest shift from writing rules to learning from data
- D. A requirement imposed by funding agencies

20. A headline reads: “AI solves protein folding.” Which pair of questions gets you furthest fastest?

- A. How much energy did it use, and where was it trained?
- B. Who funded it, and how many parameters does it have?
- C. Is it open source, and who wrote the paper?
- D. What exactly did it do and in what world, and would it survive a benchmark a rival can run?

4 Answers

#	Answer	Why
1	A	Toy success is weak evidence about the full problem. Point 4.
2	C	The 1969 proof was about the single-layer version, and backpropagation was the way round. Point 5.
3	D	A dormant field and a dead field look identical from outside. Point 1.
4	B	2012 shows that supply, not novelty, usually explains the timing. Points 8 and the headline questions.
5	A	Rules plus brute-force search, closed world, zero transfer. Point 9.
6	B	Architecture existed in 1986; compute arrived in 2007 and data in 2009. Point 8.
7	A	Perfect on what it studied, much worse on new examples: memorising, not learning. Point 8.
8	C	The field advances by benchmarks a rival can run. Point 3 and headline question 2.
9	A	Once a machine can do it, we stop calling it intelligence. Point 7.
10	D	The knowledge acquisition bottleneck; machine learning's wager is to learn it from examples. Point 6.
11	D	Cost and data are two of the four walls; the turns are science and lean. Point 10.
12	C	Engine or interface is exactly this question. Headline question 4.
13	C	Calibrated uncertainty, argued. Point 11.
14	B	Symbolic writes the rules; connectionist learns from examples. Point 3.
15	B	Architecture 1986, compute 2007, data 2009, and only the combination produced the jump. Point 8.
16	D	Pool or ocean: what world was this actually done in? Headline question 1.
17	A	Old ideas on new supply, with almost nothing new in the maths. Point 8.
18	A	Every partial combination stalls; the jump is in the combination. Point 8 and the lab box.
19	C	Camouflage and honest signalling at the same time. Point 7.
20	D	What exactly, in what world, and would a rival's benchmark confirm it. Headline questions 1 and 2.

Scored under 15? Re-read the points for the questions you missed, then try again tomorrow. The answers say which point each question came from.